

22CH203 – ENGINEERING CHEMISTRY II

UNIT V & NUCLEAR REACTIONS

1. RADIOACTIVE HALF LIFE PERIOD

Each radioactive nuclide has a characteristic, constant half-life ($t_{1/2}$), the time required for half of the atoms in a sample to decay. An isotope's half-life allows us to determine how long a sample of a useful isotope will be available, and how long a sample of an undesirable or dangerous isotope must be stored before it decays to a low-enough radiation level that is no longer a problem.

For example, cobalt-60, an isotope that emits gamma rays used to treat cancer, has a half-life of 5.27 years (Figure 1). In a given cobalt-60 source, since half of the Cobalt nuclei decay every 5.27 years, both the amount of material and the intensity of the radiation emitted is cut in half every 5.27 years. Note that for a given substance, the intensity of radiation that it produces is directly proportional to the rate of decay of the substance and the amount of the substance. Thus, a cobalt-60 source that is used for cancer treatment must be replaced regularly to continue to be effective.

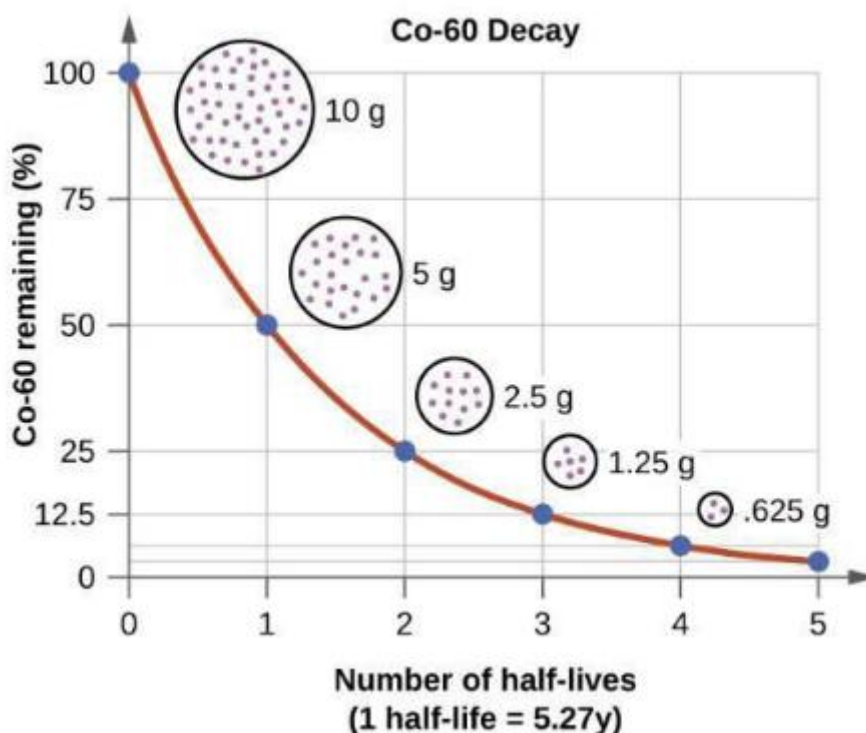


Figure 1 – Co-60 Decay

For cobalt-60, which has a half-life of 5.27 years, 50% remains after 5.27 years (one half-life), 25% remains after 10.54 years (two half-lives), 12.5% remains after 15.81 years (three half-lives), and so on. Note that every half-life is the same length of time.

Since every half-life for a radionuclide is the same length of time, we can use the following equation to calculate how much radioactive nuclide is remaining after the passage of any number (n) of half-lives:

$$\text{Isotope Remaining} = \left(\frac{1}{2}\right)^n \times \text{Starting Material}$$

• Where n = the number of half-lives determined

SAMPLE PROBLEM

If there are 60 grams of Np-240 taken initially, how much Np-240 will remain after 4 hours? (Np-240 has a half-life of 1 hour)

Solution

n= number of half-lives = 4

Isotope remaining = $(1/2)^4 \times 60$ grams = 3.75 g

After 4 hours, only 3.75g of our original 60g sample would remain

1.1. APPLICATIONS AND EFFECTS OF RADIOACTIVITY

1.1.1 APPLICATIONS IN INDUSTRY

Radioactive isotopes have found many applications in industry, both in research and in process control. In most cases, the radioactivity of tracer merely serves to define the location of a material with which it is associated. There are numerous problems that have been solved in this manner where conventional methods are valueless. Other uses of radioactive nuclides depend on the scattering of radiations active they emit. Some of the industrial applications of radiotracer in industries are as follows.

- (i) The wear of surfaces e.g. of piston rings and of gears in engines, and its prevention by means of suitable lubricants have been studied in the following manner. Steel piston is exposed to neutrons in a nuclear reactor, so that it becomes radioactive. The piston ring is then fitted into the cylinder of an internal combustion engine which is operated in the normal way with particular lubricating oil. By determining the radioactivity removed by the oil, the event of wear of the piston can be determined.
- (ii) Metals in various forms, particularly as alloys of several elements are frequently subjected to different treatment such as age-hardening, annealing, quenching, and cold rolling. In order to understand the effects of such treatments, it is desirable to know what is happening to the constituent elements, radioactive isotopes of these elements provide a simple and effective tool for following their behavior. The location of the radioactive species can be determined at any time during the treatment if the alloy by

making a radioautograph.

- (iii) The phenomenon of self-diffusion in a metal i.e. the movement of the atoms of a metal within the crystal lattice is determined with the help of radioactive traces. A block of metal, such as copper, consisting of the stable isotopes, has a layer of the same metal contain a radioisotope deposited on it by electroplating. After subjecting the block to heat treatment, stress etc, successive thin layers are shared off the surface and their activity measured. In this way, the penetrations or diffusion of the radioactive atom into the stable metal can be determined.
- (iv) Radioactive tracers also provide a convenient means for detecting leaks, especially in buried pipes carrying water or petroleum. A small quantity of radioactive substances is dissolved in the liquid near the point of the suspected leakage. The actual location of the leak can then be found by means of a sensitive gamma ray counter.
- (v) Applications In Medicine: Radioisotope have been used in the understanding of metabolic reactions, the mechanism of the action of drugs, the location of cancerous growth, obstruction in the flow of blood etc. Radioisotopes duly sterilized and carefully formulated to be safe for use an humans, are described as radiopharmaceuticals.

The following are some uses of radioisotopes in medical sciences.

- (a) Circulation of blood—to examine the restricted circulation of blood, a small quantity of sodium chloride solution in which sodium has been labelled with sodium-24 is injected into vein of the patient's forearm. A gamma-ray counter is then placed in contact with one of the feet. If the blood circulation is normal, the presence of radioactivity is much detected in foot; it increases rapidly and reaches a maximum value within less than an hour. If there is a circulatory impairment of some kind, the radioactivity with increase slowly, showing that the blood has difficulty in reaching the foot. By moving the counter to different parts of body the position of the restriction can be located and the necessary treatment can be applied.
- (b) Goitre (Thyroiditis)—Thyroid gland plays an important role in growth and metabolism. As against normal thyroid, the gland in some people becomes overactive (hyperthyroiditis) while in some other it becomes under-active (hypothyroiditis), both conditions are indicative of unhealthy. Light dose of radioactive iodine 131 in the form of NaI is given to the patient and y-activity of thyroid gland in measured. The ratio of counts Dummy/patient is measured as compared with normal hyper and hypothyroidites. Both patients are treated by radioiodine.
- (c) Brain Tumour Location—Brain tumours are difficult to locate. It is known that certain dyes fluoresceins, Rose Bengal are preferentially absorbed by cancerous cells. The technique is to label the dye with 131I as diodofluorescein, or Rose Bengal



and scan the entire space around the skull by counters. By this way it may be possible to locate the tumour fairly closely.

(d) Treatment of blood abnormalities—Phosphorus-32, the radioisotope, in the form of a solution of sodium phosphate in water is used in the treatment of some blood abnormalities, e.g. chronic leukemia, polycythemia. The radiophosphorus is taken up by the bone marrow where the red blood cells and some of the white cells are produced, the β -particle, emitted by the phosphorus-32 then inhibit the excessive formation of their cells.

1.1.2 EFFECTS OF RADIATION EXPOSURE

There is a large difference in the magnitude of the biological effects of nonionizing radiation (for example, light and microwaves) and ionizing radiation, emissions energetic enough to knock electrons out of molecules (for example, α and β particles, γ rays, X-rays, and high-energy ultraviolet radiation) Energy absorbed from nonionizing radiation speeds up the movement of atoms and molecules, which is equivalent to heating the sample. Although biological systems are sensitive to heat (as we might know from touching a hot stove or spending a day at the beach in the sun), a large amount of nonionizing radiation is necessary before dangerous levels are reached. Ionizing radiation, however, may cause much more severe damage by breaking bonds or removing electrons in biological molecules, disrupting their structure and function. Radiation can harm either the whole body (somatic damage) or eggs and sperm (genetic damage). Its effects are more pronounced in cells that reproduce rapidly, such as the stomach lining, hair follicles, bone marrow, and embryos. This is why patients undergoing radiation therapy often feel nauseous or sick to their stomach, lose hair, have bone aches, and so on, and why particular care must be taken when undergoing radiation therapy during pregnancy.

DISCUSSION QUESTIONS:

- 1. Assess the potential risks and benefits of using radioactive isotopes in medical diagnostics and treatment considering both short-term and long-term effects of radiation exposure.**
- 2. In what ways half-life influences the management and disposal of nuclear waste, and how might this affect environmental safety protocols?"**
- 3. Find the measures to minimize radiation exposure while ensuring the efficiency of radioactive applications.**